

Enhanced Thermal Human Detection with Fast Filtering for UAV Images

Umut Ozyurt

Department of Computer Engineering
Middle East Technical University
Ankara, Turkey
umut.ozyurt@metu.edu.tr

Zafer Dogan Budak

Department of Electrical and Electronics Engineering
Middle East Technical University
Ankara, Turkey
zafer.budak@metu.edu.tr

Begum Cicekdag

Department of Computer Science
Columbia University
New York, USA
bc2975@columbia.edu

Seyda Ertekin

Department of Computer Engineering
Middle East Technical University
Ankara, Turkey
sertekin@metu.edu.tr

Abstract—Real-time human detection systems can facilitate search and rescue (SAR) operations by expediting the identification of individuals in disaster-stricken areas. Unmanned aerial vehicles (UAVs), equipped with thermal imaging, can help locate missing or injured people, especially at night or in adverse weather conditions. In this study, a preprocessing filter for increasing the detection accuracy of YOLOv8 in thermal images, has been developed. Combining Contrast Limited Adaptive Histogram Equalization (CLAHE) with color value remapping filters, the proposed method creates a 3-channel RGB image from a grayscale thermal image. Experimental results show that the proposed filtering method is successful in decreasing both the false positive and the false negative rates. In addition, having minimal computational and memory requirements, it provides an alternative technique for future studies involving real-time detection in edge computing devices.

Keywords—Human detection, thermal images, infrared images, YOLOv8, unmanned aerial vehicles

I. INTRODUCTION

Search and rescue (SAR) operations that occur in the wake of natural disasters are highly time-sensitive and require rapid deployment of resources to the affected areas. Traditionally, these operations have relied heavily on human efforts. However, recent technological advancements, such as computer vision and drone surveillance, hold the potential to automate these processes, leading to the rescue of a greater number of individuals [1].

Allowing quick and easy access to disaster-stricken areas, Unmanned Aerial Vehicles (UAVs) are known for their cost-efficiency, ease of operation and maintenance, and affordability [2]. In SAR-UAV applications, RGB cameras usually face issues such as image distortion and sensitivity to changing environmental conditions, which prompts the exploration of thermal cameras for reliable human detection [3]. Infrared (IR) thermal cameras are able to capture heat emitted or reflected by objects and convert this energy into relative temperature mappings, creating single channel (gray-scale) images.

Consequently, thermal images exhibit different characteristics compared to visible light images, such as lower resolution, increased noise, and narrower dynamic ranges [4]. In Fig. 1, the distinction between visible light (RGB) and thermal images is showcased. Even though the human presence is challenging to identify in the visible light image, it is evident in the thermal capture due to the heat emitted by the individual.

While previous research in the literature has primarily focused on object detection in visible light images, there is a growing interest in thermal images with applications in autonomous driving [5], cancer detection [6], and SAR [7]. One of the most commonly used models in object detection tasks is “You Only Look Once” (YOLO), characterized by its speed and ability for real-time applications with high levels of accuracy. Unlike classifier-based systems, it “frames object detection as a regression problem to spatially separated bounding boxes and associated class probabilities” [8]. For example, Dong et al. use YOLOv3-MobileNet for real-time survivor detection in SAR scenarios [9]. Their method prunes the yolo-head layers of the neural network to decrease the required computing capacity and memory. In another study



Fig. 1: Presence of a human occluded by a tree in RGB vs thermal

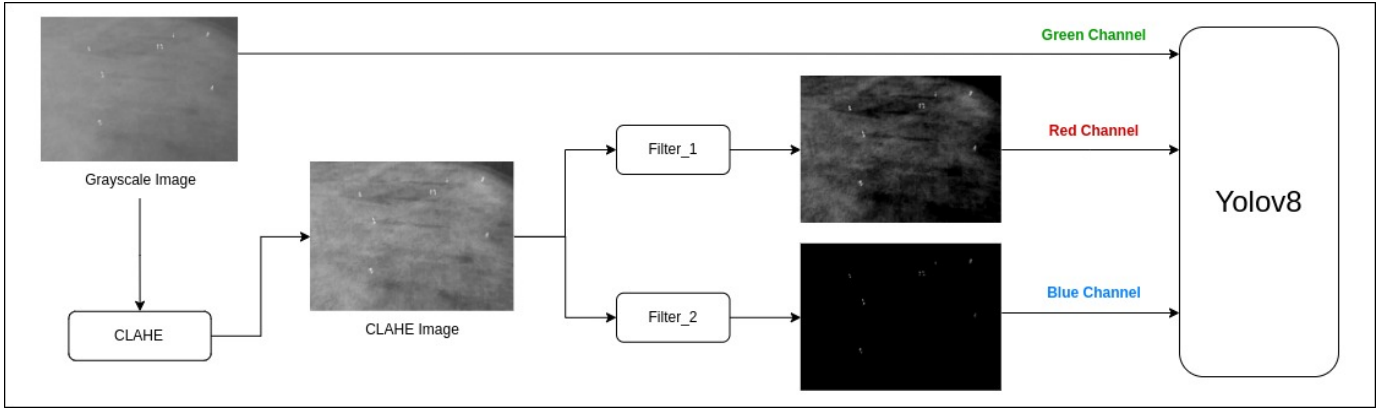


Fig. 2: Proposed pipeline

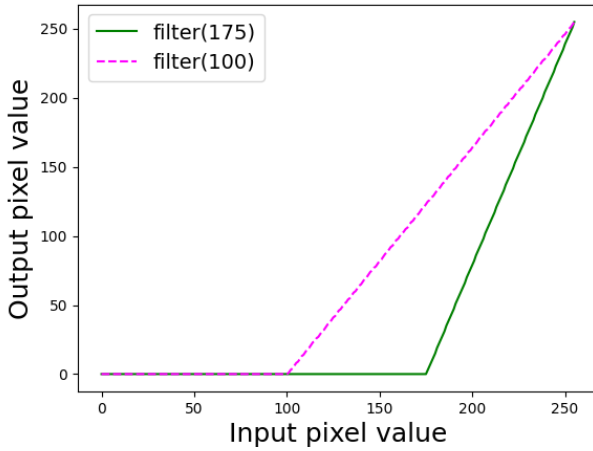


Fig. 3: Mappings of the selected custom filters

by Langman et al., the total number of people in a frame is outputted by using YOLOv5 as the detection model [10]. Various techniques such as transfer learning [11], contrastive learning [12], and transformers [13] have also been combined with YOLO to enhance its object detection capabilities. For aerial human detection, methods such as Faster R-CNN and SSD have also been used [14].

A majority of well-established object detection models, including YOLO, are designed for RGB inputs. Previous studies on thermal images have worked around this requirement by inputting the same grayscale image to all 3 channels. Unlike traditional methods, in a study by Ghose et al., saliency maps are used to obtain 3-channel versions of thermal images [15]. Through combining original and filtered versions of a thermal image, it is possible to increase the accuracy of object detection models. However, the practicality of these current techniques on edge devices such as drones remains low due to the potential extended inference time.

In this study, a simple yet effective method for human detection during night-time SAR operations has been proposed. Thermal images obtained from these scenarios are unique as

the individuals in the frames have a higher contrast than their backgrounds. In order to leverage this distinction, the proposed method uses a combination of Contrast Limited Adaptive Histogram Equalization (CLAHE) [16] with two custom color value filters. Enabling the object detection model to focus on human presence, a 3-channel RGB image from a single channel thermal image is created. The processed images along with the original one are inputted to distinct channels. For detection, a YOLOv8-nano [17] model is trained. Evaluations are performed using a curated dataset that combines thermal images from HIT-UAV [18] and NII-CU datasets [19].

II. PROPOSED METHOD

The architecture of the proposed method can be found in Fig. 2. The grayscale thermal images are initially subjected to CLAHE, which increases the contrast of the image. This histogram equalization step further darkens the background while brightening the humans. The processed image is then passed through a custom function, which is explained in detail in Section 2.2. Once the pre-processing is completed, the images are inputted into the RGB channels. The original image is inputted into the green channel. The image obtained through CLAHE and the mapping function with a starting point of 100, as an 8-bit pixel value, is inputted into the red channel. The input to the blue channel goes through the same steps as the red except for the starting point, which is set to 175. The inputs are then combined and fed to the YOLOv8 [17] model for training.

A. Contrast Limited Adaptive Histogram Equalization

CLAHE is a local contrast enhancement algorithm for pre-processing images. Through constraining amplification, CLAHE mitigates noise-overamplification, which can occur in regular adaptive histogram equalization (AHE). The detailed CLAHE procedure is as follows:

- 1) The input image is partitioned into smaller, non-overlapping tiles or regions. This is determined by the tile grid size inputted by the user.
- 2) A histogram is computed for each tile, representing the pixel distribution within that region.

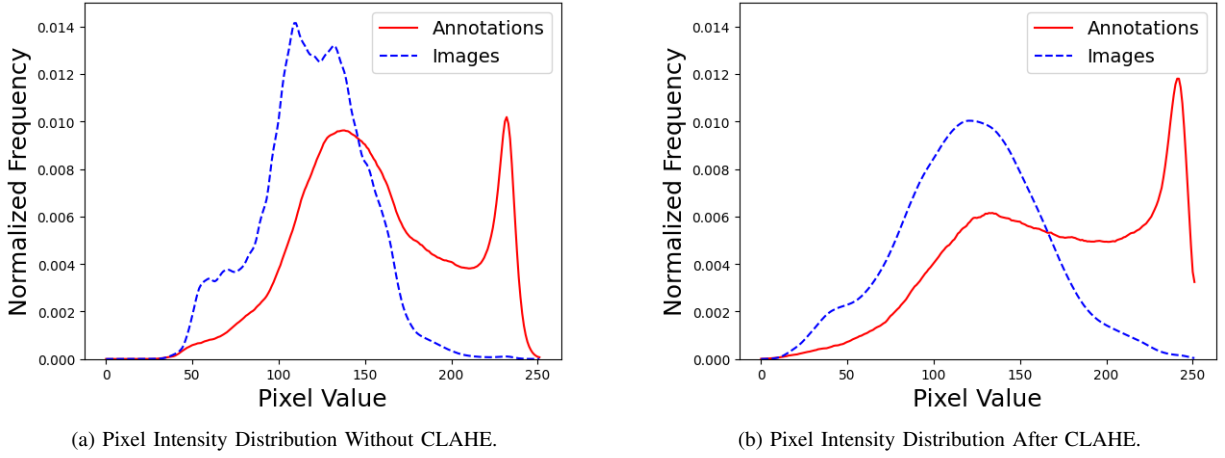


Fig. 4: Pixel distributions of the dataset images and annotations with and without CLAHE

- 3) Histogram bin values higher than the clip limit are accumulated and distributed into other bins. This control mechanism ensures that the enhancement remains visually pleasing while effectively improving the overall contrast of the image.
- 4) A cumulative distribution function (CDF) is calculated for the histogram.
- 5) CDF values of each tile are scaled and mapped using the input image pixel values.
- 6) The resulting tiles are stitched together using bilinear interpolation to generate an output image with improved contrast.

At night, the human body is typically warmer than its immediate environment. This temperature contrast causes humans to appear brighter and whiter in thermal images. This whiter color indicates that humans emit more infrared radiation (heat) than the background. Using the CLAHE filter makes this distinction even more pronounced, making it easier for the model to distinguish between individuals and other irrelevant objects in the background. The CLAHE filter used in this study has the "Clip Limit" parameter as 2 and the "Tile Grid Size" as 4x4.

B. Custom Filter

Let $\text{filter}(s)$ be the pixel mapping function defined as:

$$\text{filter}(s) = \begin{cases} 0 & \text{if } x < s \\ \frac{255(x-s)}{255-s} & \text{if } s \leq x \leq 255 \end{cases}$$

where s is the starting point and x is the input pixel value in the range starting from 0 to 255. The 2 filter variants of these mappings can be observed in Fig. 3.

This approach aims to focus on the brighter (hotter) pixel values. Despite these custom filters' resemblance to ordinary thresholding methods, they do not output binary values since the differences between the pixels are valuable, in contrast, they furthermore enhance these pixels' value differences. Since the probability of encountering a human presence is higher in brighter areas in the image, two filters with different purposes

are used to feed the red and blue channels in the YOLOv8 model, while the green channel is reserved for the original image to prevent any information loss. The experiments were done with two filters: one filter with a starting point that is relatively low (100) and one filter with a higher starting point (175). The main purpose of the higher, which means more strict, filter is to force the model to focus on the areas where all the possibilities of any human presence might exist. In short, we designed it to increase the recall value. The ultimate goal of the lower (softer) filter is to increase both recall and precision by catching a balance between preserving the necessary background details and highlighting humans. Our main goal was to increase the number of human detections without making the filters too expensive in terms of computational speed to maintain similar frame per second (FPS) values as it was in the normal inferences. On NVIDIA Jetson Xavier NX, the filter takes on average of 1.5 ms for 640x640 images in single-core computation.

C. YOLOv8 Detection

The last step of the proposed pipeline is the YOLOv8-nano model. We have chosen this model, one of the most recent object detection models, for two main reasons. First of all, YOLOv8-nano is fast and lightweight enough to be used in edge device deployments [20] including SAR operations, even with the small and relatively cheap devices such as NVIDIA Jetson Xavier NX. Secondly, YOLOv8 models are superior compared to their YOLO ancestors due to the cutting-edge and robust approaches used in its architecture. Even when it is trained with not filtered images, it achieves successive results for small human detections in the evaluation metrics, which can be seen in Fig. 5

By the time this paper was submitted to be reviewed, the official YOLOv8 paper had yet to be released. However, the GitHub repository was released in January 2023 by Ultralytics, the publisher of one of the former YOLO versions, the YOLOv5 [21]. YOLOv8 is based on convolutional neural networks (CNN) designed for computer vision tasks such as

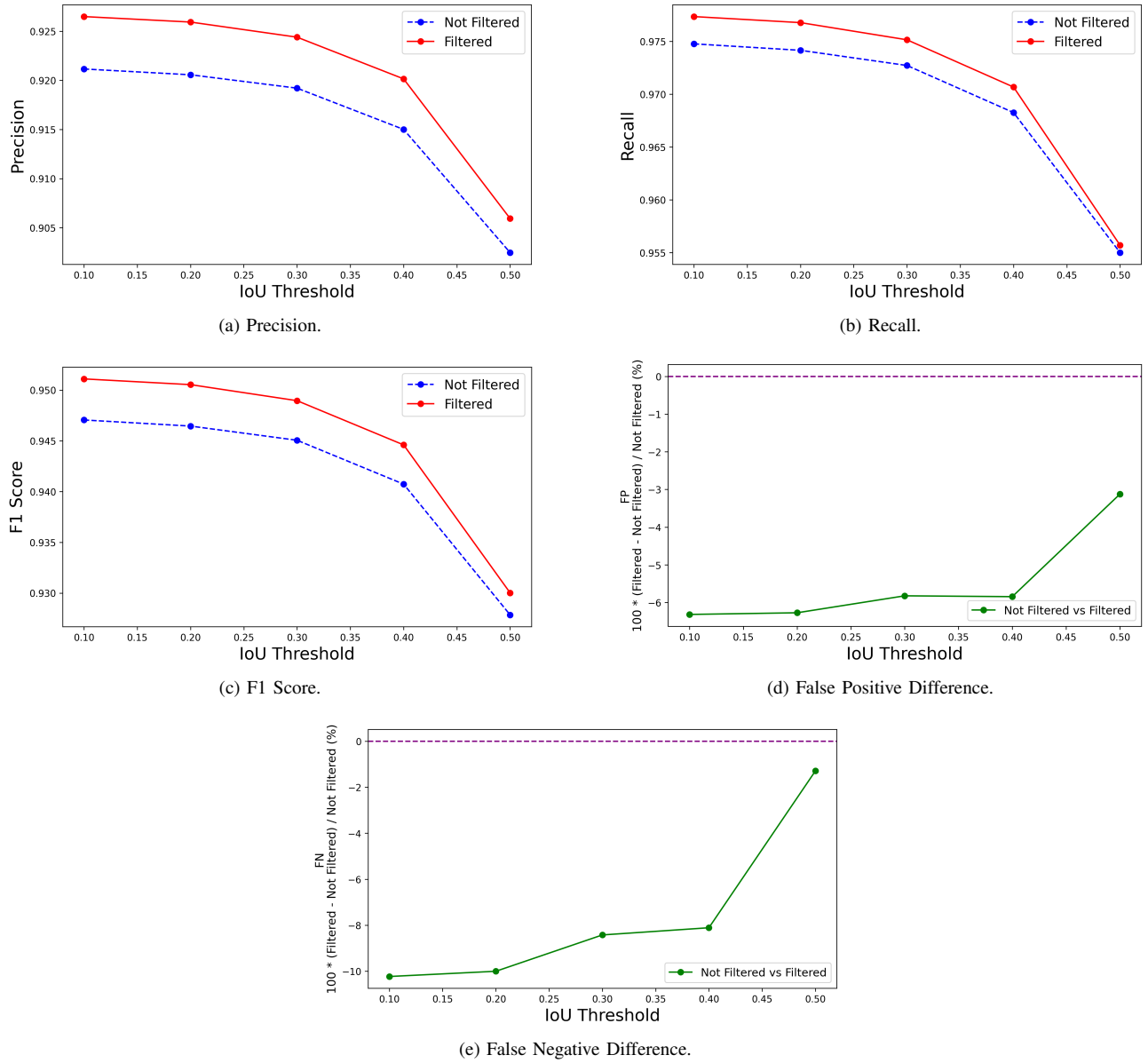


Fig. 5: Comparisons of metrics between filtered and not filtered

classification, segmentation, pose estimation, object tracking, or object detection, as it was used in this study. It consists of three main parts as its older versions.

The backbone part extracts features, the neck part connects the backbone network to the head network, and finally, the head outputs final predictions. The backbone basically recognizes high-level features such as edges, colors, or basic patterns since it has the initial layers of the YOLOv8 network [22]. This part also contains "shortcut connections" as it was introduced in the res-net [23] study. The neck part is effective in keeping the balance between the speed and the accuracy tradeoff. After the head part, the outputs are ready after the non-maximum suppression (NMS) algorithm, which is a post-processing technique. The YOLOv8 is still heavily maintained by the open-source community, and it is getting

better day by day. Hence, it can be considered as open-to-improve for future applications, which is essential for its life cycle.

Similar to its ancestor YOLOv6 [24], the YOLOv8 has anchor-free detection. Using the anchor boxes is a technique where pre-defined rectangles exist for each grid cell in the model's output. The main intention is to perform better at the images where two or more distinct objects exist with close enough center points. In these cases, anchor boxes provide flexibility to detect the objects separately according to their shapes, which causes the output to be much larger and the NMS algorithm slower. The NMS technique is used to eliminate multiple detection boxes for the same object, iterating through every detection candidate. Speeding up this process saves some computation time for YOLOv8 for its other

features. Moreover, the advantage of using anchor boxes is that they will only be effective if there is a possibility of very close overlapping objects by their center positions, which is not the case for most SAR operations. Furthermore, the deployment phase of YOLOv8 is considered convenient since it is easy to export to various model types to deploy on edge devices.

III. RESULTS AND DISCUSSION

The proposed method was evaluated using a custom thermal image dataset that can be accessed through Roboflow [25]. It contains a total of 3411 ground truth humans across 654 images selected from the HIT-UAV [18] and NII-CU datasets [19]. During night-time SAR operations, thermal imaging is more desirable than RGB as the latter generally produces dark or indistinct images. At night, humans continue to emit heat while the ambient temperature drops, meaning that humans have a higher contrast than the background. For the purposes of this study, images that carry the characteristics of daytime thermal images were mostly removed from both datasets. Mis-labeled or significantly similar images were also excluded to ensure dataset quality.

For the human detection task, 20 YOLOv8-nano [17] models were trained for 150 epochs with a batch-size of 16. In order to build the models from the YAML configuration file, the model parameter was specified as "yolov8n.yaml" in the training prompt. Ten models were trained on ten randomly shuffled versions of the original dataset and ten models for the matching filtered pairs of these datasets. Mean averages are taken over these 10 runs for the filtered and not-filtered variations. The dataset was distributed such that the training, validation, and test subsets comprised 70%, 20%, and 10% of the entire dataset, respectively.

When determining the starting points of the blue and red mappings, an experimental approach was used. Optimal results were obtained when the values were set to 175 for blue and 100 for red. The Hue, Saturation, and Value Augmentations (HSV) were not utilized. In aerial imagery, objects can appear in different spatial distributions, scales, and orientations. The mosaic augmentation was set to 0.2 to expose the model to the above conditions during training.

The pixel distributions of images pre- and post-CLAHE can be found in Fig. 4. Fig. 4b shows that a majority of annotations, which correspond to humans, have a pixel value between 225-255 when CLAHE is used. The humans in equalized images are white, while the backgrounds have pixel values between 100-150. This significant contrast allows the model to detect objects easily compared to the raw pixel distribution, where the values for the annotations and the image as a whole overlap.

The confidence level threshold was set to 0.2 when identifying object detections. For performance measurements, varying values of Intersection over Union (IoU) were used. IoU quantifies the overlap between predicted and ground-truth bounding boxes and can be calculated as

$$IoU = \frac{|A \cap B|}{|A \cup B|} \quad (1)$$

where A is the area of the predicted bounding box and B is the area of the ground-truth bounding box. The IoU threshold determines which predictions can be counted as True Positives. For SAR operations, identifying the presence or absence of individuals carries more importance than the exact location of said individuals. Therefore, high IoU thresholds where the predicted coordinates match the ground truth ones with a high overlap are not required. The IoU thresholds used for evaluations are 0.1, 0.2, 0.3, 0.4, and 0.5.

Experiment results can be found in Fig. 5, which displays the improvements in the confusion matrix values as well as in metrics such as precision, recall, and F1 score. Across all metrics and IoU thresholds, the proposed method outperforms the traditional with notable decreases in the False Positive and False Negative rates. Fig. 6c displays an image where the proposed method was successful in identifying a human that wasn't otherwise identified. In SAR scenarios, identifying all missing or injured individuals is crucial. Since false negatives are the undetected humans, reducing this rate is critical. An image exemplifying this scenario can be seen in Fig. 6a. On the other hand, the false positives are the objects other than humans that have been falsely detected. Such incorrect identifications can waste the rescue team's time when swift and efficient responses are required. The decrease in false positive and false negative rates, calculated as the average of the evaluation IoU values, are 5.5% and 7.6%, respectively.

IV. CONCLUSION

Improving the effectiveness of SAR missions through object detection and real-time situational awareness, computer vision has become increasingly important in disaster management. Unmanned aerial vehicles (UAVs), equipped with thermal imaging, are characterized by their ability to quickly cover broad areas and provide visibility at night or in adverse weather conditions. Through human detection models that are able to accurately identify individuals, SAR operations can be significantly expedited. Due to its speed, accuracy and real-time capabilities, YOLO is one of the most widely used models in the computer vision field. Designed for detection in visible light images, YOLO accepts inputs from three channels: red, blue, and green.

In this study, the performance of YOLOv8 is increased by preprocessing grayscale thermal images and inputting distinct versions of the original image to each channel. Combining CLAHE with color value remapping filters, the proposed method creates a 3-channel RGB image from a single channel thermal image. Experimental results show that the proposed method is especially successful in decreasing the false positive and false negative rates with slight improvements in precision, recall, and the F1 score. Locating all individuals in the affected zones is the desired outcome in SAR operations. Therefore, minimizing false negatives, which represent individuals that go undetected, is essential. In contrast, false positives refer to objects, aside from humans, that have been mistakenly identified. These detections can cause unnecessary deployment of resources.

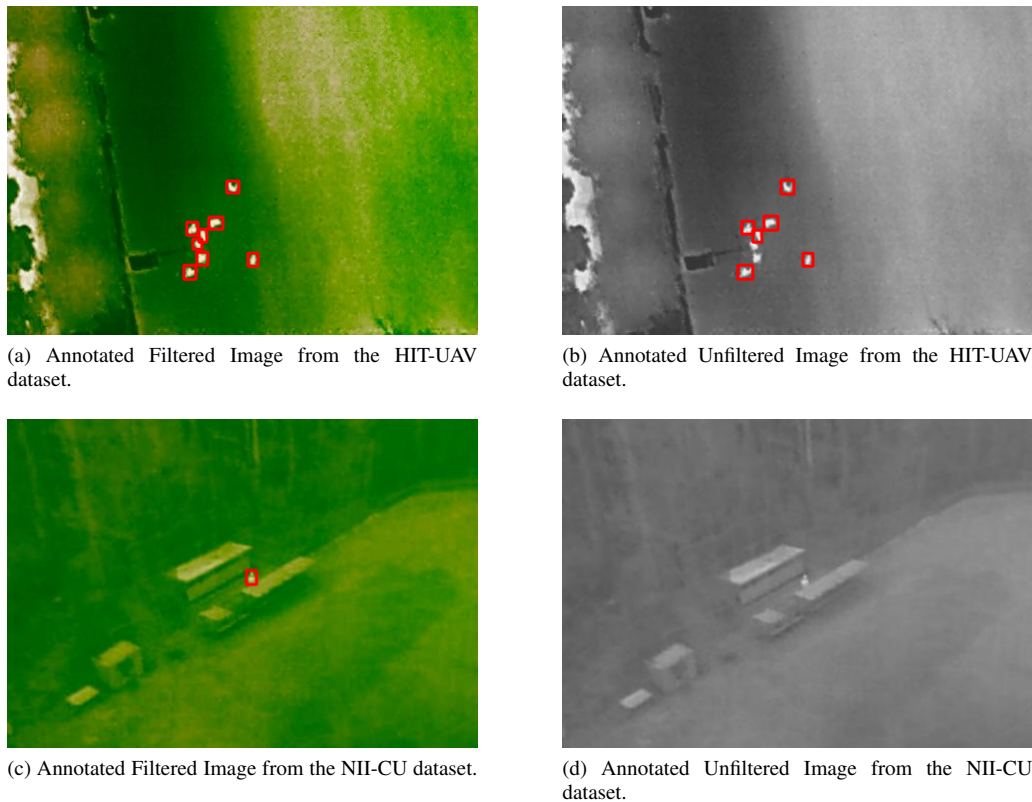


Fig. 6: Annotated bounding boxes as predicted by the proposed and traditional methods.

In conclusion, the proposed method is a straightforward but effective technique that can be used for human or object detection tasks in thermal images. Its low computational and memory demands make it well-suited for integration with other object detection models as well as other preprocessing techniques.

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